Reading the Reading Room

Pneumonia classification from chest X-rays

Three baseline experiments, with declared reconstruction limits
Lung cropping with a CNN and ViT ensemble proposal

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Introduction and project objective

Task

Classify each pediatric chest X-ray as normal or pneumonia.

M2 objective

Reconstruct the published ensemble and report its actual errors.

M3 objective

Propose lung cropping and a pretrained ViT, with separate controls.

Literature reviewed in M1

Paper	Main idea
P1 Kermany 2018	Pediatric CXR dataset and transfer learning
P2 CheXNet 2017	DenseNet for chest X-ray classification
P3 Wang 2017	Large chest X-ray dataset and localization benchmarks
P4 Stephen 2019	Train a compact CNN from scratch
P5 Ayan and Unver 2019	Compare VGG16 and Xception
P6 Rahman 2020	Compare pretrained CNN backbones
P7 Chouhan 2020	Five CNNs combined by majority vote SELECTED
P8 Rajaraman 2018	CNN classification and activation-map interpretation
P9 Singh 2024	Pretrained DeiT-Base Vision Transformer

Selected baseline and reconstruction protocol

P7 Chouhan et al. 2020

AlexNet, DenseNet121, InceptionV3, ResNet18 and GoogLeNet
Frozen early layers, trained late blocks, three-of-five vote

Paper partition	Normal	Pneumonia	Total
Train + original val	1,349	3,883	5,232
Original test	234	390	624

The 15-epoch audit held out 1,042 images and trained on 4,164.
The corrected five-epoch run uses all 5,232 with fresh image augmentation.
Paper schedules are 50 to 200 epochs. Author settings remain partly unknown.

Three runs and their measured results

Run	Epochs	Train	Accuracy	Precision	Status
Initial cached	5	5,232	87.18%	83.12%	Complete
Cached audit	15	4,164	85.74%	81.55%	Complete
Online correction	5	5,232	92.15%	92.95%	Verified

Run 3 removes feature caching and samples new augmentation each epoch.
It restores the paper training partition and uses fixed epoch-five weights.

624 test records for every measured row. Crop and ViT remain untested.

Final results compared with the paper

Metric	Paper reported	Our final 5 epochs
Accuracy	96.39%	92.15%
Pneumonia precision	93.28%	92.95%
Pneumonia recall	99.62%	94.62%
ROC AUC	99.34%	96.23%

Precision is close to the paper. Accuracy is 4.24 percentage points lower.

Our test: 624 images. Our training: 5 epochs per model. Paper: 50 to 200.
Several author settings remain unspecified, so this is a partial reproduction.

M3 proposed crop and CNN ViT system

Lung-focused input, complementary classifiers

1 Crop both lungs with a 10% margin
More input resolution for lung tissue, less peripheral variation

2 Combine the five CNNs with a pretrained small ViT
Patch self-attention may contribute different predictions

Two changes, separate hypotheses. Crop is the primary M3 change.
Weights and thresholds use validation only. Both improvements remain untested.

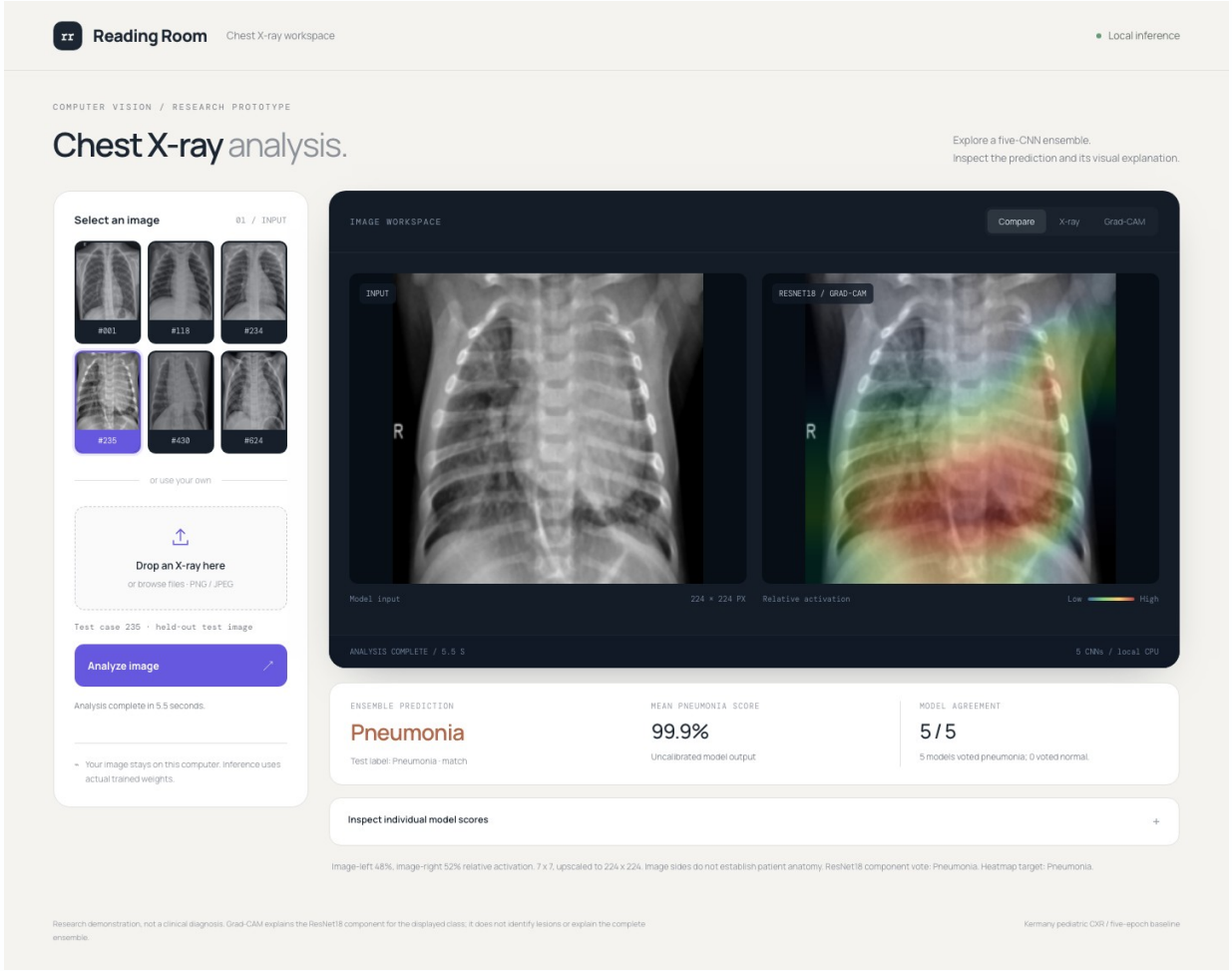
Comparisons and prediction before implementation

Run	Image region	Classifiers
A	Full X-ray	Five CNNs
B	Lung crop	Five CNNs
C	Full X-ray	CNNs plus pretrained ViT
D Final proposal	Lung crop	CNNs plus pretrained ViT

Combined prediction versus A

Macro F1 +0.01 to +0.04. Accuracy 0 to +3 percentage points.
Pneumonia recall loss no more than 2 points. Gains need not add up.

Live demo and next experiment



Real inference

Test samples or upload
Five trained CNN votes
ResNet18 Grad-CAM

Next: crop plus ViT

Four matched comparisons
No new crop or ViT results
JEPA and mixup later

Local demo: 127.0.0.1:8765. Demo uses original CNN weights. No clinical claim.